

Figure 12.3 **(**a**)** Conventional theory is based on electrons and quarks as the basic constituents of matter. **(**b**)** String theory suggests that each particle is actually a vibrating string.

Here is where string theory makes its appearance. String theory challenges the conventional picture by proposing that electrons and quarks are *not* zero-sized particles. Instead, the conventional particle-as-dot model, according to string theory, is an approximation of a more refined portrayal in which each particle is actually a tiny, vibrating filament of energy, called a *string*, as you can see in Figure 12.3b. These strands of vibrating energy are envisioned to have no thickness, only length, and so strings are one-dimensional entities. Yet, because the strings are so small, some hundred billion billion times smaller than a single atomic nucleus (10^{-33} centimeters), they appear to be points even when examined with our most advanced atom smashers.

Because our understanding of string theory is far from complete, no one knows for sure whether the story ends here—whether, assuming the theory is correct, strings are truly the final Russian doll, or whether strings themselves might be composed of yet finer ingredients. We will come back to this issue, but for now we follow the historical development of the subject and imagine that strings are truly where the buck stops; we imagine that strings are *the* most elementary ingredient in the universe.

String Theory and Unification

That’s string theory in brief, but to convey the power of this new approach, I need to describe conventional particle physics a little more fully. Over the past hundred years, physicists have prodded, pummeled, and pulverized matter in search of the universe’s elementary constituents. And, indeed, they have found that in almost everything anyone has ever encountered, the fundamental ingredients are the electrons and quarks